



SYNTHETIC FINANCIAL FRAUD DETECTION

Advanced Fraud Detection Powered by Synthetic Data & Machine Learning.

PROBLEM STATEMENT



🛡 Rising online financial fraud threatens global banking systems daily

💳 Traditional rule-based systems fail to detect evolving fraud patterns



⚠ ML DETECTS FRAUD FASTER

Machine Learning models analyze millions of transactions in real time, identifying suspicious patterns that traditional systems miss with greater speed and accuracy.

🔒 IMPROVES TRANSACTION SECURITY

Advanced ML algorithms strengthen digital payment security, reducing false positives and protecting customers from synthetic and real-world financial fraud.

OBJECTIVES

✓ Secure Digital Payments
Apply robust ML pipelines to safeguard online and digital banking transactions against fraud.

✓ Real-Time Fraud Prediction
Deploy a live prediction system via web application enabling instant fraud detection on new transactions.



✓ **DETECT FRAUDULENT TRANSACTIONS**

Identify and flag suspicious financial transactions in real-time using ML classification models.

✓ **IMPROVE PREDICTION ACCURACY**

Maximize model precision and recall to minimize false positives and missed fraud cases.

✓ **REDUCE FINANCIAL LOSSES**

Prevent unauthorized transactions early to protect users and institutions from monetary damage.

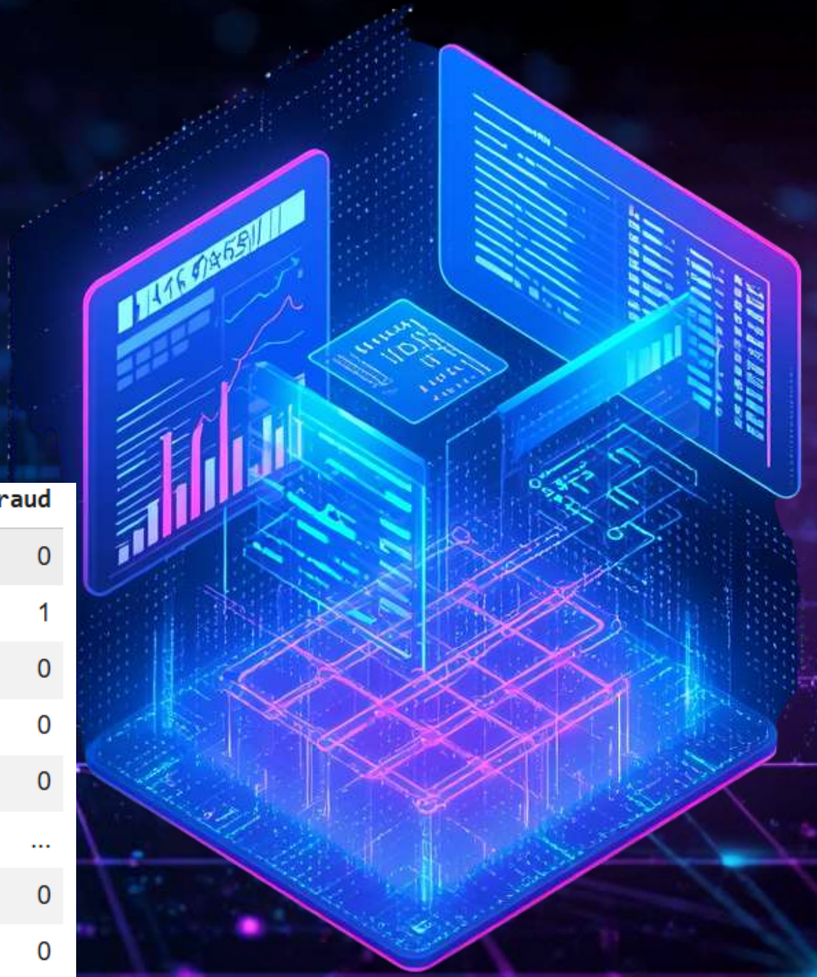
DATASET OVERVIEW

SYNTHETIC FINANCIAL DATASET

TRANSACTION RECORDS & INPUT FEATURES

	transaction_id	amount	transaction_type	account_age_days	num_prev_transactions	location_risk	device_type	is_fraud
0	1	46.93	cash_out	816	249	high	desktop	0
1	2	301.01	debit	1474	362	high	mobile	1
2	3	131.67	transfer	454	440	medium	desktop	0
3	4	91.29	debit	1948	20	medium	desktop	0
4	5	16.96	debit	629	195	low	tablet	0
...	...	...	...	...	...	...	...	...
995	996	9.61	payment	1252	183	high	mobile	0
996	997	249.27	cash_out	1720	285	medium	tablet	0
997	998	14.71	payment	1643	56	high	tablet	0
998	999	300.05	transfer	596	221	low	desktop	0
999	1000	59.06	debit	1874	302	medium	mobile	0

1000 rows × 8 columns



FRAUD & GENUINE CLASSES — BINARY CLASSIFICATION

Target variable: isFraud (0 = Genuine, 1 = Fraudulent). Enables supervised binary classification using ML algorithms to detect anomalies.

DATA PREPROCESSING PIPELINE

A structured preprocessing pipeline transforms raw financial transaction data into a clean, model-ready format — ensuring accuracy, consistency, and optimal performance for fraud detection.



⚙ STEP 1: DATA CLEANING & VALIDATION

Raw dataset loaded → Missing values handled → Duplicate records removed → Data types validated and corrected for consistency.

⚙ STEP 2: ENCODING, SCALING & SPLITTING

Label Encoding applied to categorical features → StandardScaler normalizes numeric values → Train-Test Split (80/20) prepares data for model training.

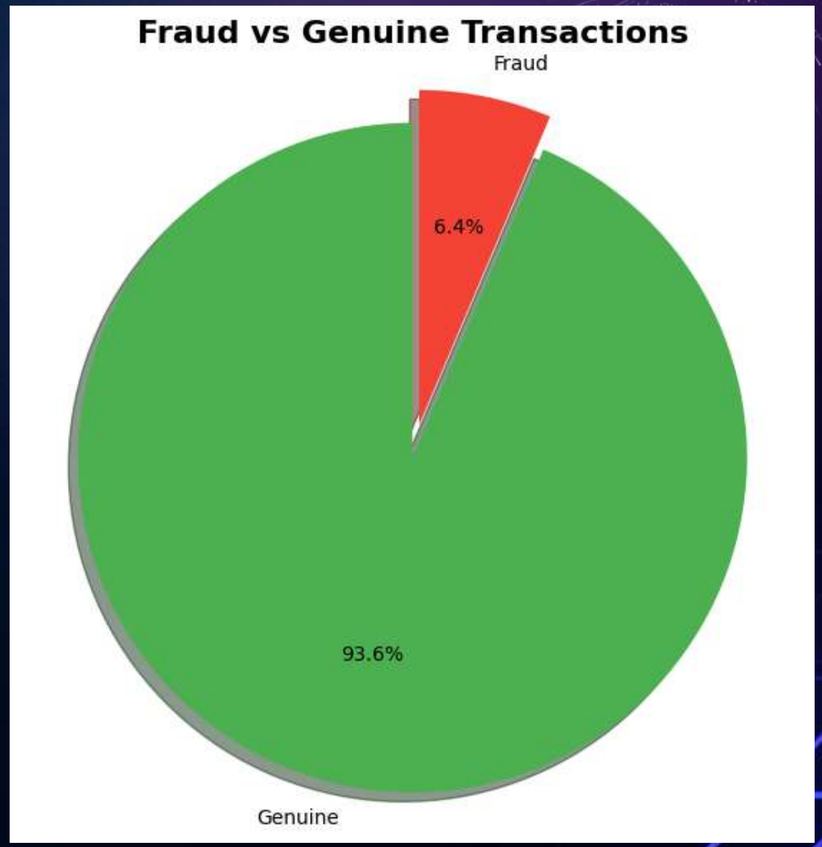
EXPLORATORY DATA ANALYSIS

Visual analysis of transaction data distribution to understand fraud vs. genuine class imbalance and proportional breakdown across the synthetic data set. Charts below illustrate the frequency and proportion of fraudulent vs. legitimate transactions, revealing key class distribution patterns critical for model training.

📊 COUNT PLOT — TRANSACTION DISTRIBUTION



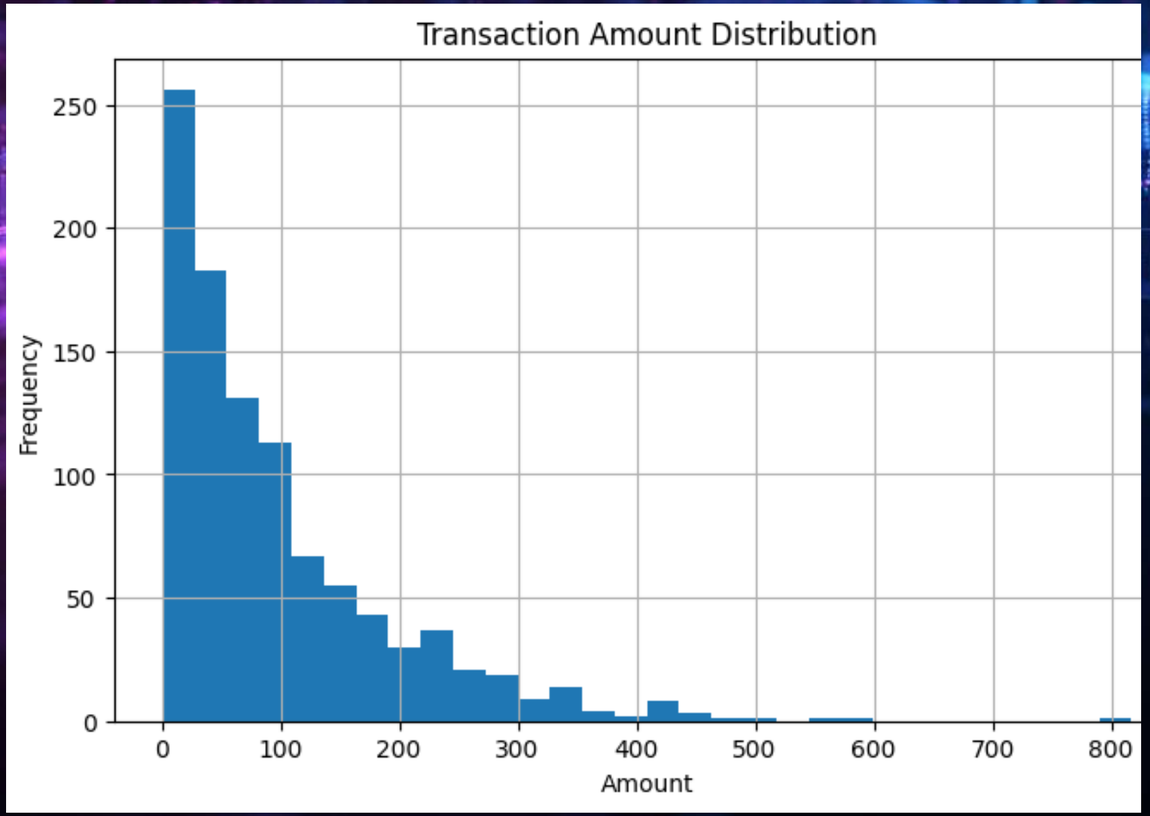
🥧 PIE CHART — FRAUD VS. GENUINE RATIO



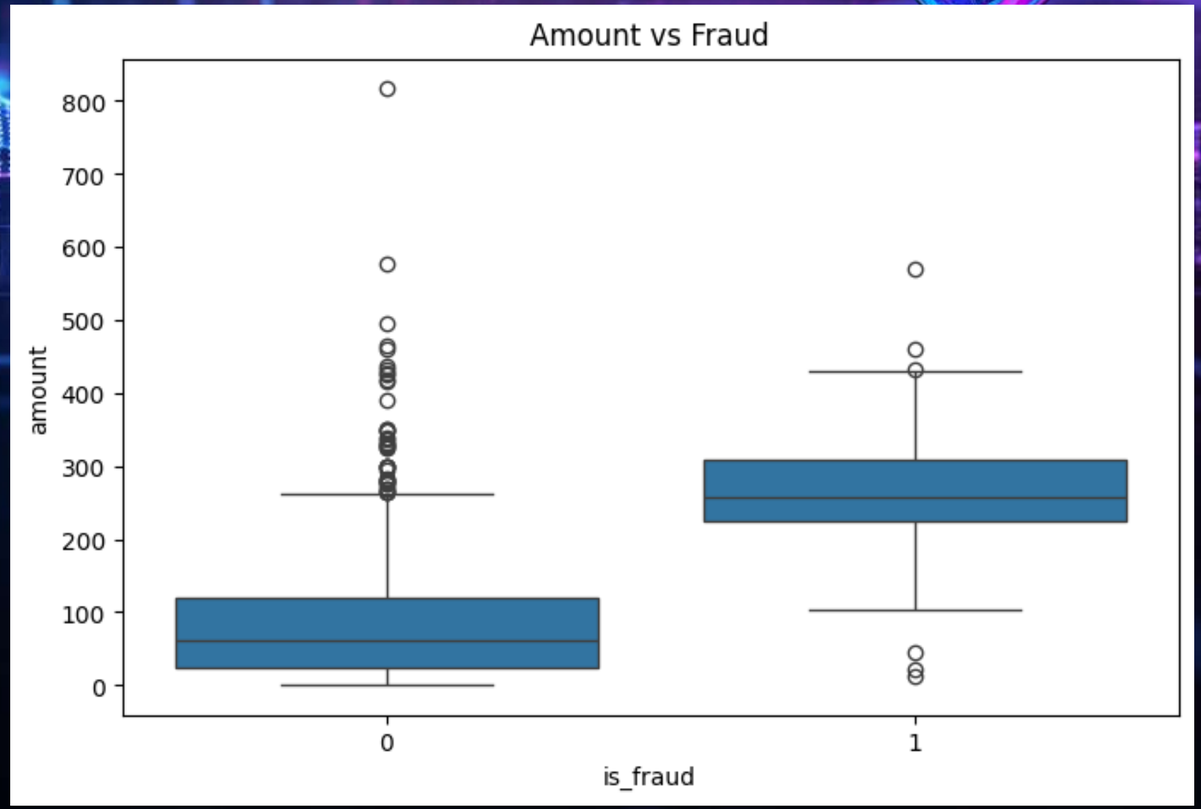
DATA ANALYSIS

Visual exploration of transaction amount distributions and feature spread across fraud and genuine classes.

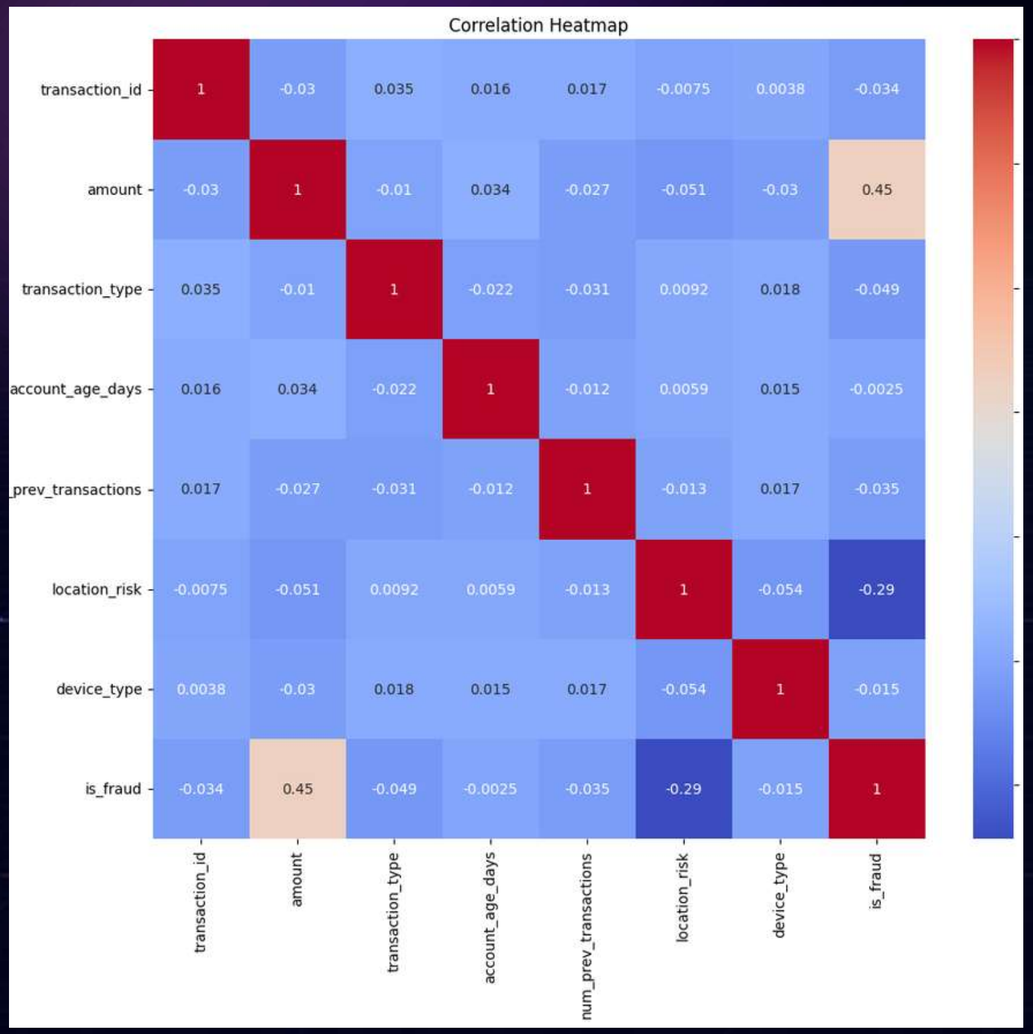
HISTOGRAM
TRANSACTION AMOUNT DISTRIBUTION



BOX PLOT
FEATURE SPREAD & OUTLIERS



CORRELATION ANALYSIS



FEATURE CORRELATION HEATMAP

Key Insight: Strong positive correlations observed between transaction amount, balance changes, and fraud labels — confirming these as critical predictive features.

MACHINE LEARNING MODELS

An ensemble learning method that builds multiple decision trees and merges their results. Handles imbalanced data, resistant to overfitting, and delivers high accuracy for fraud detection tasks.

⋆ RANDOM FOREST (MAIN MODEL)

Logistic Regression: A baseline linear classifier using probability thresholds. Fast and interpretable but struggles with complex non-linear fraud patterns.
Decision Tree: Rule-based splits for classification. Easy to visualize but prone to overfitting on high-dimensional transaction data.

LOGISTIC REGRESSION & DECISION TREE

Key Metric: F1 Score | Precision | Recall
All models evaluated using cross-validation on synthetic financial dataset with binary fraud classification target.

XGBOOST — GRADIENT BOOSTING

XGBoost: Extreme Gradient Boosting with regularization. Highly competitive accuracy, handles missing values, and excels in large-scale financial fraud datasets.



MODEL PERFORMANCE

★ ACCURACY & PRECISION

Accuracy: 0.965— *Overall correct predictions out of total transactions evaluated.*

Precision: 1.00 — Ratio of true fraud detections among all flagged transactions.

RECALL & F1 SCORE

Recall: 0.53 — *Proportion of actual fraud cases correctly identified by the model.*

F1 Score: 0.6 — Harmonic mean of Precision and Recall; balanced performance metric.



FRAUDSHIELD AI WEB APPLICATION



HOME PAGE

Clean landing interface displaying the FraudShield AI dashboard. Features navigation menu, fraud detection summary stats, and a call-to-action button to begin transaction analysis.

PREDICTION FORM

Interactive input form where users enter transaction details — amount, type, account balances — and submit data to the Random Forest ML model for real-time fraud prediction.

PREDICTION RESULT

Displays the ML model's verdict: Fraudulent or Genuine. Shows confidence score, flagged risk indicators, and recommended action in a clear, color-coded result card.

SYSTEM ARCHITECTURE

1 MODEL DEVELOPMENT PIPELINE

fraud_dataset.csv
Synthetic Financial Dataset

Data Preprocessing
- ✓ Data Cleaning
- ✓ Missing Value Handling
- ✓ Duplicate Removal

Feature Engineering
- ✓ Label Encoding
- ✓ Feature Scaling (StandardScaler)
- ✓ Feature Selection

Train-Test Split
80% Training
20% Testing

Machine Learning Models
Logistic Regression · Decision Tree · KNN · Random Forest (BEST MODEL)

Model Training & Evaluation
Accuracy, Precision, Recall, F1-Score
Confusion Matrix, ROC-AUC

Save Model
best_model.pkl
scaler.pkl

best_model.pkl
(Trained Random Forest Model)

2 WEB APPLICATION (DEPLOYMENT PIPELINE)

User
Enters Transaction Details (Amount, Type, Age, etc.)

HTML Interface
User Inputs via Web Form

CSS Styling
Beautiful & Responsive UI

Flask Backend (app.py)
Receives Input & Processes Request

Load Model
Load best_model.pkl & scaler.pkl

Prediction Engine
Random Forest Model Predicts the Transaction

Legitimate Transaction
(Not Fraud)

Fraudulent Transaction
(Fraud Detected)

DATASET FEATURES (Input Variables)

Transaction ID	Amount	Transaction Type	Account Age	Previous Transactions	Location Risk	Device Type	Target (is_fraud)
Unique ID of Transaction	Transaction Amount (USD)	Payment, Transfer, Cash Out, etc.	Account Age (in Days)	Number of Previous Transactions	Low, Medium, High Risk	Mobile, Laptop, Desktop, Tablet	0 = Legitimate 1 = Fraudulent

KEY BENEFITS

- High Accuracy Fraud Detection
- Real-Time Prediction
- Secure & Scalable System
- Easy Integration
- Helps Prevent Financial Losses

WORKFLOW & TECHNOLOGY

Transaction Data → Preprocessing → Feature Engineering → Random Forest Model → Fraud Prediction → Web Application Output

ML PIPELINE WORKFLOW

- Python
- Flask
- Pandas
- NumPy
- HTML & CSS

TECHNOLOGY STACK

Random Forest serves as the core prediction engine, integrated with a responsive Flask web app for real-time fraud detection and user-friendly result display.

FRAUDSHIELD AI ARCHITECTURE

End-to-end ML pipeline from raw transaction input to real-time fraud prediction via a Flask-powered web interface.





THANK YOU